

Lab Assignment 3.

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Project Overview

1. The *Library Manager* is a lightweight, command line application designed to help campus libraries to manage thier book catalog efficiently. It streamlines tasks such as maintaining book records, tracking issued/returned books, managing users, and monitoring fines and due dates.
2. Purpose of the Project The goal of this project is to simplify library operations by digitizing the entire workflow. The system ensures:
 - a) Easy management of books
 - b)Secure tracking of members.
 - c) Fast issue and return processes.
 - d) Reduced human errors.
 - e)Data consistency and integrity.
3. It uses Object-Oriented Programming(OOP) principles and *JSON file persistence* to keep records structured, reliable and easy to maintain.

HOW TO USE THE PROGRAM:

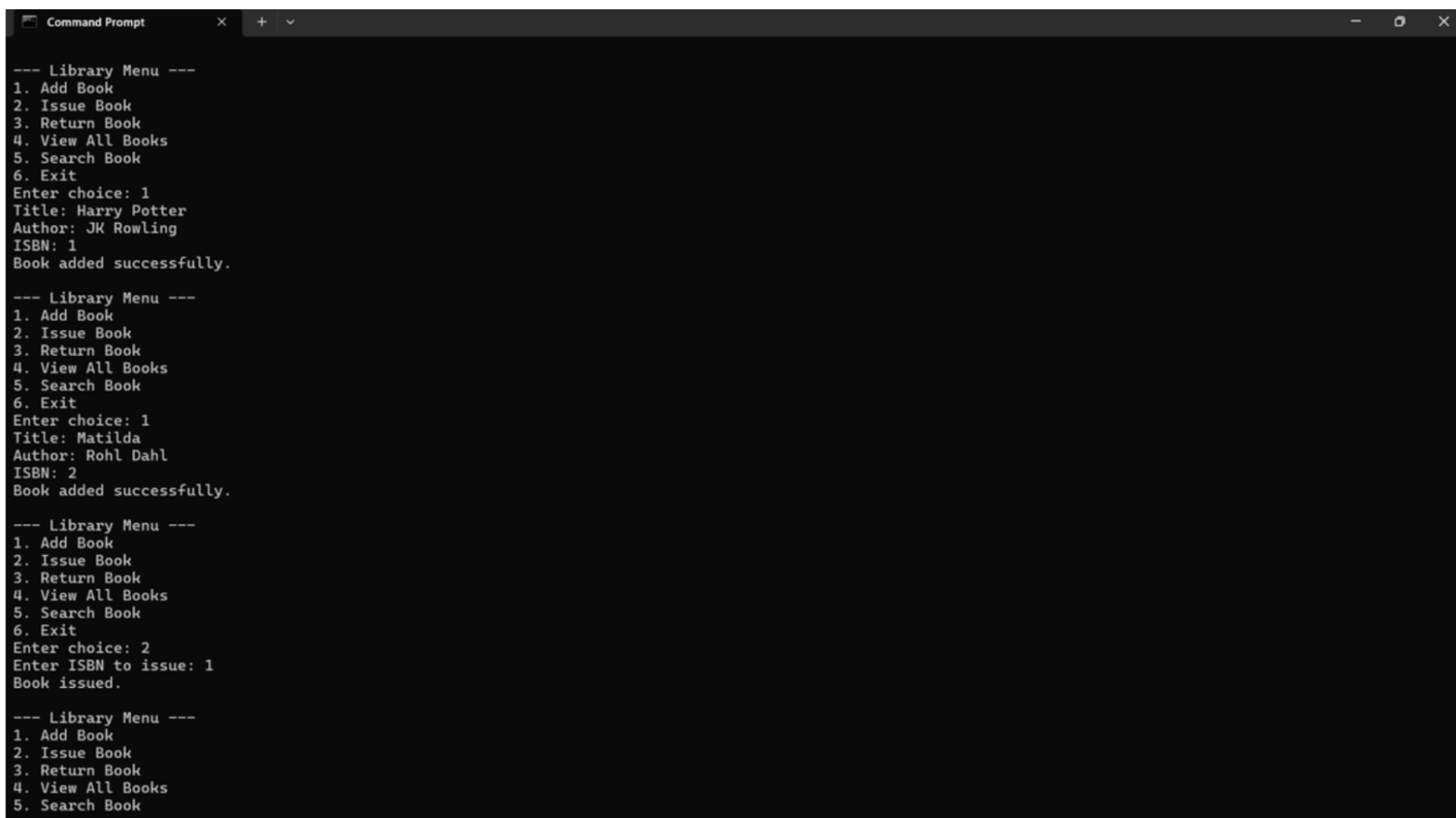
1. Run the cli python- climain.py
2. Choose an option from the menu.

```
--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice:
```

3. Data Storage:

- a) All records are saved in data/ catalog.json
- b) if the file is missing or corrupted the system automatically resets and creates a new one.

OUTPUT SCREENSHOT:



```
Command Prompt
--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 1
Title: Harry Potter
Author: JK Rowling
ISBN: 1
Book added successfully.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 1
Title: Matilda
Author: Rohl Dahl
ISBN: 2
Book added successfully.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 2
Enter ISBN to issue: 1
Book issued.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
```

```
Command Prompt
Book issued.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 2
Enter ISBN to issue: 1
Book not available.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 3
Enter ISBN to return: 1
Book returned.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 4
Harry Potter by JK Rowling | ISBN: 1 | Status: available
Matilda by Rohl Dahl | ISBN: 2 | Status: available

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 5
Enter title: 2
```

```
Command Prompt
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 4
Harry Potter by JK Rowling | ISBN: 1 | Status: available
Matilda by Rohl Dahl | ISBN: 2 | Status: available

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 5
Enter title: 2
No book found.

--- Library Menu ---
1. Add Book
2. Issue Book
3. Return Book
4. View All Books
5. Search Book
6. Exit
Enter choice: 6
Exiting...

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